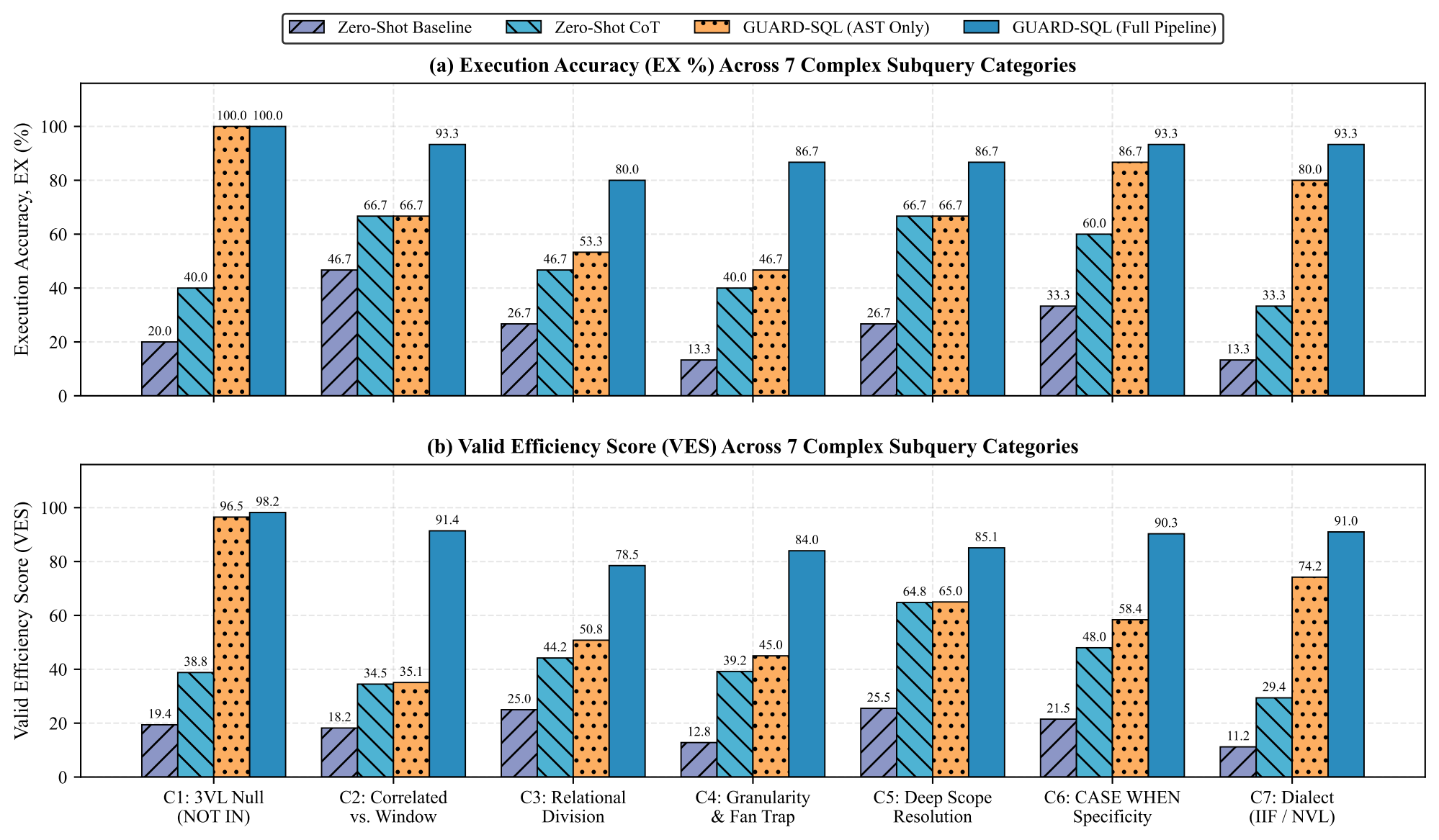

Zero-Shot Baseline
Zero-Shot CoT
GUARD-SQL (AST Only)
GUARD-SQL (Full Pipeline)
(a) Execution Accuracy (EX %) Across 7 Complex Subquery Categories
Execution Accuracy, EX (%)
0
20
40
60
80
100
20.0
40.0
100.0
100.0
46.7
66.7
66.7
93.3
26.7
46.7
53.3
80.0
13.3
40.0
46.7
86.7
26.7
66.7
66.7
86.7
33.3
60.0
86.7
93.3
13.3
33.3
80.0
93.3
(b) Valid Efficiency Score (VES) Across 7 Complex Subquery Categories
Valid Efficiency Score (VES)
0
20
40
60
80
100
19.4
38.8
96.5
98.2
18.2
34.5
35.1
91.4
25.0
44.2
50.8
78.5
12.8
39.2
45.0
84.0
25.5
64.8
65.0
85.1
21.5
48.0
58.4
90.3
11.2
29.4
74.2
91.0
C1: 3VL Null (NOT IN)
C2: Correlated vs. Window
C3: Relational Division
C4: Granularity & Fan Trap
C5: Deep Scope Resolution
C6: CASE WHEN Specificity
C7: Dialect (IIF / NVL)